

UNIVERSITÄT
LEIPZIG

International Physics Studies Program

Advanced Physics Laboratory

Alpha Particle Spectrometry

with a Semiconductor Detector

Group: O4

Date: 12 May 2026

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Abstract

Alpha-particle spectrometry with a silicon surface-barrier detector was used to investigate the energy spectrum of a ^{241}Am source and the energy loss of alpha particles in air and Mylar. First, the dependence of peak width and count rate on source-detector distance was studied under vacuum. Slot 6 was selected as a suitable compromise between energy resolution and counting statistics. A calibration spectrum recorded at this position gave a fitted peak channel of $\mu = 3341.03 \pm 0.17$ and a calibration factor of $a = (1.64187 \pm 0.00008)$ keV/channel using the dominant ^{241}Am alpha energy of 5485.56 keV. The detector energy resolution was found to be $R = 0.0064 \pm 0.0001$, corresponding to a FWHM of (34.88 ± 0.74) keV. A SRIM simulation of a 50 nm silicon dead layer gave a mean energy loss of 6.65 keV and a straggling contribution of 4.57 keV. The alpha-particle range in air was determined by measuring the peak energy at different source-detector distances and extrapolating $E_{\max}$ to zero. The experimental range was $R_{\exp} = (41.02 \pm 0.22)$ mm, in good agreement with the empirical prediction of 40.86 mm and the SRIM average range of 40.46 mm. Energy-loss measurements in Mylar showed a decrease in transmitted energy from (5482.32 ± 0.76) keV without Mylar to (811.21 ± 11.65) keV at 29 μm. The extrapolated Mylar range was approximately 31.50 μm, differing from the SRIM value of 32.751 μm by 3.81%. Overall, the measurements reproduce the expected stopping behavior of alpha particles and show good agreement with SRIM simulations within the experimental limitations.

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1. Introduction

Alpha-particle spectrometry is a useful method for studying both nuclear decay energies and the interaction of charged particles with matter. In this experiment, alpha particles emitted by a ^{241}Am source are detected with a silicon semiconductor detector. Since alpha decay produces particles with well-defined kinetic energies, the measured spectrum can be used to calibrate the spectrometer and to investigate how alpha particles lose energy while passing through different media. The most probable alpha-particle energy of ^{241}Am is used as the calibration reference, with $E_\alpha = 5485.56$keV [1]. Nuclear data for decay schemes and radiation data are commonly tabulated in databases such as NuDat/ENSDF, which provides evaluated nuclear-structure and decay information [2].

The main physical process studied in this experiment is the slowing down of alpha particles in matter. As positively charged heavy particles travel through a medium, they interact primarily with atomic electrons, causing excitation and ionization. These interactions reduce the kinetic energy of the alpha particles and lead to a finite range in the material. The stopping power describes the energy loss per unit path length and is one of the central quantities used to characterize alpha-particle slowing in matter. For heavy charged particles, the mean energy loss is described conceptually by Bethe-type stopping theory, where the stopping power depends strongly on particle velocity and material properties [3].

A characteristic feature of alpha-particle energy loss is that the stopping power increases as the particle slows down. Therefore, the energy loss is not constant along the path. Close to the end of the particle range, the energy deposition rises strongly, producing a Bragg-curve-like behavior. This is why alpha particles are strongly ionizing but have a relatively short penetration depth. The NIST ASTAR database provides stopping-power and range tables for helium ions in materials including dry air and polyethylene terephthalate/Mylar, which are the two relevant absorber media in this experiment [4].

Experimentally, the alpha particles were measured using a silicon surface-barrier detector. When an alpha particle enters the active detector region, it creates electron-hole pairs. The charge carriers are separated by the electric field in the depletion region, producing a voltage pulse whose height is proportional to the deposited energy. The pulse-height distribution was then recorded as a channel spectrum by the multi-channel analyzer. The detector arranged as a silicon semiconductor detector mounted above a sample chamber with adjustable slots for source and absorber positioning [1].

After calibration, the range of alpha particles in air was determined by measuring spectra at different source-detector distances in air and extrapolating the measured peak energy $E_{\max}$ to zero. The resulting range was compared with the empirical air-range relation and with SRIM simulations. The experimental results were compared with SRIM/TRIM simulations. The experimental results are finally compared with SRIM/TRIM simulations in order to assess the measured ranges and stopping powers.

Overall, this experiment connects nuclear spectroscopy with charged-particle transport in matter. The measured spectra are used not only to identify the alpha-particle energy, but also to quantify how this energy changes with air path length and absorber thickness. The final comparison with empirical range formulas and SRIM simulations provides a check of how well the experimental setup reproduces the expected stopping and range behavior of alpha particles in air and Mylar.

2. Theoretical Background

2.1 Nuclear Decay and Alpha Radiation

Alpha decay is a radioactive decay process in which an unstable nucleus emits a helium nucleus, consisting of two protons and two neutrons. In this experiment, the alpha source is ^{241}Am, which decays mainly into ^{237}Np by emitting alpha particles:

$$^{241}_{95}\mathrm{Am} \rightarrow {}^{237}_{93}\mathrm{Np} + {}^{4}_{2}\mathrm{He}^{2+} + Q. \tag{1}$$

The decay energy Q is shared between the emitted alpha particle, the recoiling daughter nucleus, and possible excitation energy of the daughter nucleus. Because the daughter nucleus can be left in different excited states, the alpha particles are not emitted with only one single energy. Instead, several discrete alpha energies are possible. For the calibration in this experiment, the most probable alpha line of ^{241}Am is used:

$$E_\alpha = 5485.56\ \mathrm{keV}. \tag{2}$$

The lab manual lists this line with an emission probability of 84.8%, making it the dominant peak in the measured alpha spectrum [1]. Nuclear decay data of this type are also tabulated in evaluated databases such as NuDat/ENSDF [2].

2.2 Interaction of Alpha Particles with Matter

Alpha particles are heavy, positively charged particles. When they pass through matter, they lose energy mainly through Coulomb interactions with the electrons of the absorbing medium. These interactions excite or ionize atoms and molecules. Nuclear scattering with target nuclei can also occur, but for the energies relevant here, electronic energy loss is the dominant process.

Because alpha particles carry charge $z_\alpha = 2$, they ionize matter strongly. However, they also lose energy quickly and therefore have a short range compared with more weakly ionizing radiation. In this experiment, this behavior is studied in two media: air and Mylar. In Task 3, the distance travelled through air is varied. In Task 4, Mylar foils of

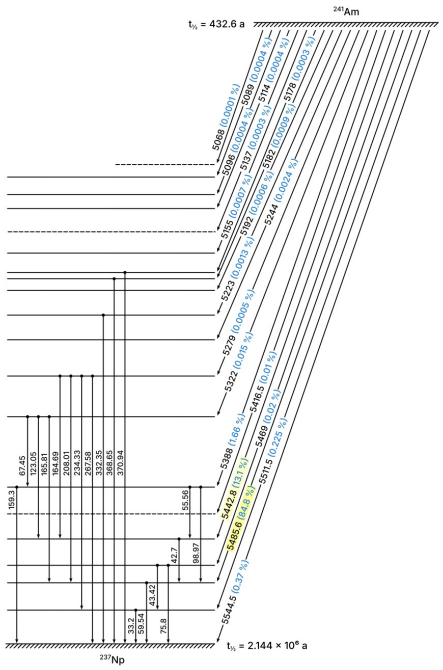


Figure 1: Term diagram of the alpha decay of ^{241}Am. The dominant alpha line at 5485.56 keV is used for energy calibration. Figure adapted from the lab manual [1].

different thicknesses are inserted between source and detector.

The energy loss of charged particles in matter is treated generally in stopping-power theory. The Particle Data Group describes the electronic energy loss of heavy charged particles in matter, including ionization, excitation, maximum energy transfer, and stopping power [3].

2.3 Stopping Power and Bragg Curve

The rate of energy loss per path length is called the stopping power:

$$S(E) = -\frac{dE}{dx}. \tag{3}$$

The minus sign is included because the kinetic energy of the alpha particle decreases as the travelled distance x increases. For charged particles, the stopping power depends on the particle charge, velocity, and the properties of the absorber material. In the Bethe-type description, the stopping power contains an approximate dependence on $1/v^2$ for non-relativistic heavy charged particles:

$$-\frac{dE}{dx} \propto \frac{1}{v^2}. \tag{4}$$

Therefore, slower alpha particles generally lose more energy per unit distance [1, 3].

This velocity dependence leads to a characteristic Bragg curve. At first, the alpha particle is fast, so the stopping power is relatively moderate. As it slows down, it spends more time interacting with nearby electrons, so the stopping power increases. Close to the end of the range, the energy deposition reaches a maximum. After the alpha particle becomes slow enough to capture electrons and become neutral helium, the stopping power drops rapidly.

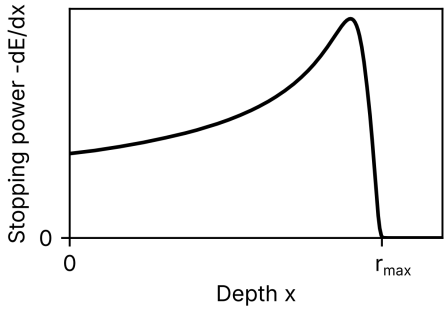


Figure 2: Schematic Bragg curve showing stopping power as a function of penetration depth. The stopping power increases as the alpha particle slows down and drops when the particle comes to rest. Figure adapted from the lab manual [1].

2.4 Range of Alpha Particles

The range of an alpha particle is the distance it travels in a medium before its mean kinetic energy is reduced to zero. Experimentally, this can be determined by measuring spectra after different absorber thicknesses or path lengths and plotting the most probable energy $E_{\max}$ against distance. The range is then obtained by extrapolating this curve to:

$$E_{\max} = 0. \tag{5}$$

For alpha particles in air, the lab manual gives the empirical relation:

$$r_{\max}(\text{cm}) = 0.318 \cdot E^{3/2}, \tag{6}$$

where E is the alpha-particle energy in MeV. This relation applies to air at 15°C and 1010 mbar [1]. For the dominant ^{241}Am alpha energy of 5.48556 MeV, this formula predicts a range of about 40.9 mm in air.

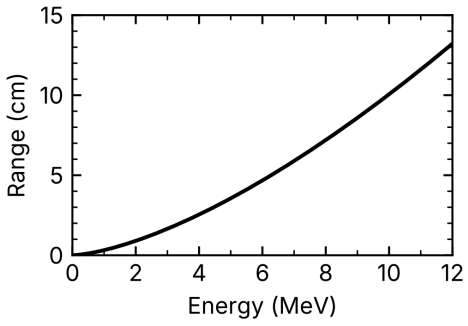


Figure 3: Empirical range of alpha particles in air as a function of kinetic energy. Figure adapted from the lab manual [1].

2.5 Energy and Range Straggling

The energy loss of alpha particles in matter is statistical. Even alpha particles with the same initial energy do not all undergo exactly the same sequence of collisions. Therefore, after passing through a material layer, the transmitted particles

have a broadened energy distribution. This broadening is called energy straggling.

Similarly, the distance at which alpha particles stop is not exactly the same for every particle. Instead, the range is distributed around a mean value. This effect is called range straggling. In this report, straggling is not evaluated through a separate range-distribution model; it is treated as one of the physical causes of peak broadening, especially at large air distances and large Mylar thicknesses.

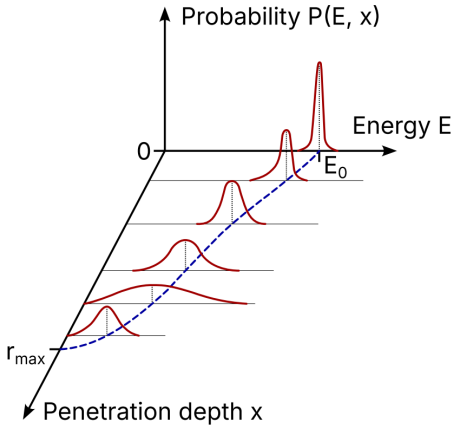


Figure 4: Broadening of the energy distribution of initially quasi-monoenergetic alpha particles as they pass through a medium. The mean energy decreases with penetration depth, while statistical fluctuations in the energy-loss process lead to energy straggling. Adapted from the lab manual [5].

Straggling is important in this experiment because it broadens the measured peaks, shown in Fig: 4. The effect becomes stronger for larger air distances or thicker Mylar foils, especially near the end of the alpha-particle range. Therefore, peaks at large distances or large Mylar thicknesses are broader and harder to fit accurately.

2.6 Semiconductor Detector and Energy Resolution

In a silicon semiconductor detector, an incident alpha particle loses energy mainly by ionizing the detector material and creating electron-hole pairs. The collected charge is proportional to the deposited particle energy, so the measured pulse height can be used as an energy signal. In silicon, the average energy required to create one electron-hole pair is about 3.62 eV, because part of the deposited energy is also transferred to lattice vibrations during thermalization [1].

A real detector does not produce infinitely sharp peaks. The width of a measured peak is described by its full width at half maximum, FWHM. The relative energy resolution is defined as:

$$R = \frac{\mathrm{FWHM}}{E_{\max}}. \tag{7}$$

The detector also has a dead layer near the entrance surface. Alpha particles lose some energy in this layer before reaching the active detector volume. Since charge produced in the dead layer is not fully collected, this contributes to peak broadening and energy straggling.

2.7 SRIM Simulations

SRIM/TRIM is used to simulate the transport of helium ions through matter. The lab manual describes SRIM as a Monte Carlo program that calculates ion trajectories, energy losses, ranges, and stopping powers using interaction cross sections for electronic and nuclear stopping [1].

In this experiment, SRIM is used for two main purposes. First, it is used to estimate the contribution of the silicon detector dead layer to the measured energy broadening. Second, it is used to simulate alpha-particle energy loss and stopping power in air and Mylar. The NIST ASTAR database provides an independent reference for helium-ion stopping powers and ranges in materials such as dry air and polyethylene terephthalate/Mylar [4].

3. Experimental Setup and Methods

3.1 Alpha-Particle Spectrometer

The experiment was performed using an alpha-particle spectrometer with a ^{241}Am radioactive source and a silicon semiconductor detector. The source and detector were mounted inside a sample chamber. The source holder could be inserted into discrete slot positions, allowing reproducible variation of the source-detector distance. The same slot system could also be used to insert absorber materials, such as Mylar foils, between the source and detector. According to the lab manual, the distance between neighboring slots is 4 mm. The chamber could be evacuated using a vacuum pump, while a pressure gauge was used to monitor the pressure. Ambient pressure could be restored using a ventilation valve [1].

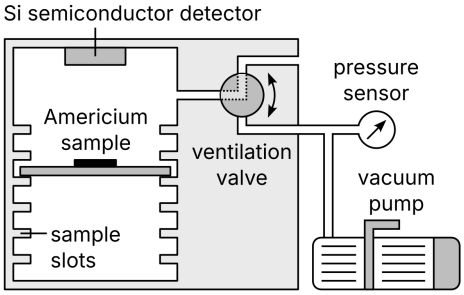


Figure 5: Experimental setup of the alpha-particle spectrometer, showing the sample chamber, silicon detector, source and absorber slots, pressure gauge, vacuum pump, and ventilation valve. Adapted from the lab manual [1].

The use of vacuum was important for Tasks 1, 2, and 4. In vacuum, energy loss in air is minimized, so changes in peak shape or energy can be attributed mainly to geometry, detector effects, or the inserted absorber. For Task 3, the chamber was intentionally operated with air in the source-detector path so that the energy loss of alpha particles in air could be measured as a function of distance.

3.2 Surface-Barrier Detector and Signal Chain

The detector used in the spectrometer was a silicon surface-barrier detector. When an alpha particle entered the active detector region, it produced electron-hole pairs by ionizing the semiconductor. The electric field in the depletion region separated these charge carriers and generated a charge pulse. The charge pulse was sent to a preamplifier, then processed by the main amplifier. The amplified pulse height was proportional to the collected charge and therefore to the deposited alpha-particle energy. The analog pulse was digitized by an analog-to-digital converter and assigned to a channel number. The multi-channel analyzer then accumulated a histogram of counts versus channel number, giving the measured alpha spectrum [1].

The recorded spectra were therefore initially obtained as count rate versus channel number. To convert channel number into energy, an energy calibration using the known ^{241}Am alpha energy was required.

3.3 Radiation Protection

Since the experiment used a radioactive ^{241}Am source, the radiation protection rules given in the manual were followed. The basic principles were to minimize the time spent close to the source, maintain distance where possible, and use shielding and storage rules appropriately. The source was not touched with unprotected hands, unnecessary manipulation of the source was avoided, and unused radioactive emitters were kept securely stored. Eating, drinking, smoking, use of cosmetics, and unnecessary touching of exposed skin were avoided during handling of radioactive material [1].

3.4 Measurement Programme

The experiment consisted of four main measurement parts. First, spectra were recorded under vacuum at different source-detector distances in order to identify a suitable compromise between energy resolution and count rate. Second, a longer calibration spectrum was recorded under vacuum using the selected geometry. Third, spectra were recorded in air for different source-detector distances to determine the alpha-particle range in air. Finally, Mylar foils of different thicknesses were placed between source and detector under vacuum in order to measure the transmitted alpha-particle energy and determine the stopping power of Mylar.

For Task 4, all Mylar spectra were recorded with the source at Slot 8. Although Slot 6 had been selected as a good compromise for the calibration measurement, the full Mylar-thickness series could not be measured mechanically at Slot 6 because the thicker foil combinations obstructed the source holder. Therefore, Slot 8 was used for all Mylar measurements, and a separate no-Mylar Slot 8 reference spectrum was used as the zero-thickness reference.

Table 1: Summary of the main experimental measurement conditions.

Task	Condition	Purpose
Distance optimization	Vacuum	Determine intensity and FWHM versus slot position
Calibration	Vacuum	Calibrate channel axis using the ^{241}Am reference energy
Air range	Air	Determine $E_{\max}$ versus source-detector distance
Mylar energy loss	Vacuum	Determine $E_{\max}$ versus Mylar thickness

3.5 Peak Fitting and Energy Calibration

The measured alpha peaks were not perfectly symmetric. In particular, the low-energy side of the peak was broadened because some alpha particles lost energy inside the source material, absorber, or detector entrance region before depositing energy in the active detector volume. Therefore, the peak position was determined by fitting the high-energy side of the main peak with a Gaussian function:

$$G(x) = A \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right] + C. \tag{8}$$

Here, A is the peak amplitude, μ is the fitted peak position, σ is the Gaussian width, and C is a constant background offset. The FWHM was calculated from the fitted width using:

$$\mathrm{FWHM} = 2\sqrt{2\ln 2}\sigma. \tag{9}$$

These two equations are part of the data evaluation method rather than additional formulas from the manual. The manual requires the FWHM and energy maximum to be determined, while the Gaussian fit provides a consistent way to extract them from asymmetric spectra.

The calibration spectrum was recorded at Slot 6 under vacuum. The fitted peak position of the dominant ^{241}Am alpha line was assigned to the reference energy:

$$E_\alpha = 5485.56\ \mathrm{keV}. \tag{10}$$

Since only one calibration line was used, the calibration was treated as a zero-intercept linear calibration. The channel width was calculated as:

$$a = \frac{E_\alpha}{\mu_{\mathrm{cal}}}. \tag{11}$$

The energy corresponding to any channel x was then obtained from:

$$E(x) = ax. \tag{12}$$

The uncertainty of the converted energy was calculated by propagating the uncertainty of the fitted channel position and the uncertainty of the channel width:

$$\sigma_E = \sqrt{(x\sigma_a)^2 + (a\sigma_x)^2}. \tag{13}$$

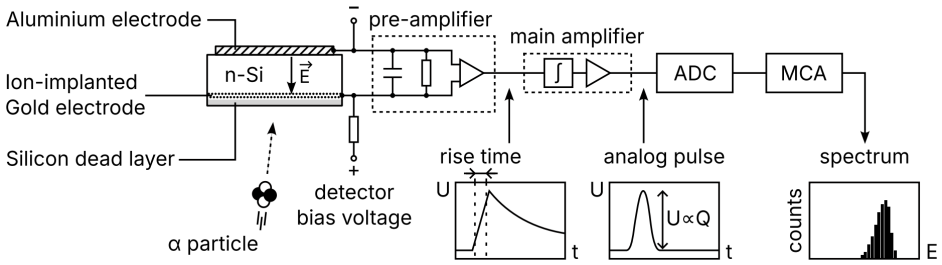


Figure 6: Principle of the silicon surface-barrier detector and associated counting electronics. The charge pulse generated by an incident alpha particle is amplified, digitized, and stored as a channel spectrum. Adapted from the lab manual [1].

3.6 Range Measurement in Air

For the air-range measurement, alpha spectra were recorded for source-detector distances from Slot 1 to Slot 10. The corresponding distances used in the analysis were:

$$d = 3,\ 7,\ 11,\ 15,\ 19,\ 23,\ 27,\ 31,\ 35,\ 39 \text{ mm}. \tag{14}$$

For each distance, the main alpha peak was fitted and converted into an energy value $E_{\max}(d)$ using the calibration. As the distance in air increased, the alpha particles lost more energy before reaching the detector, so the fitted peak energy decreased.

The range was determined by fitting the measured $E_{\max}(d)$ values with a polynomial function and extrapolating the curve to:

$$E_{\max}(r_{\max}) = 0. \tag{15}$$

The resulting experimental range was later compared with the empirical air-range formula given in the manual and with the SRIM simulation for alpha particles in air.

3.7 Energy Loss and Stopping Power in Mylar

For the Mylar measurements, spectra were recorded under vacuum for different foil thicknesses. The analyzed Mylar thicknesses were

$$x = 2.5,\ 5,\ 6,\ 7.5,\ 11,\ 12,\ 14.5,\ 17,\ 18,\ 20.5,\ 23,\ 24,\ 26.5,\ 29\ \mu\text{m}. \tag{16}$$

A no-Mylar reference spectrum was also recorded at Slot 8 and used as the zero-thickness reference. This was important because all Mylar spectra were measured at Slot 8. The energy loss caused by a Mylar thickness x was calculated as:

$$\Delta E(x) = E_0 - E(x), \tag{17}$$

where E_0 is the fitted energy of the Slot 8 no-Mylar reference spectrum and $E(x)$ is the fitted transmitted energy after the Mylar foil.

The experimental stopping power was calculated from neighboring Mylar thicknesses using a finite-difference approximation:

$$S_i = -\frac{\Delta E_i}{\Delta x_i} = \frac{E(x_i) - E(x_{i+1})}{x_{i+1} - x_i}. \tag{18}$$

Each stopping-power value was assigned to the midpoint of the corresponding thickness interval:

$$x_{i,\text{mid}} = \frac{x_i + x_{i+1}}{2}. \tag{19}$$

This finite-difference method follows directly from the definition of stopping power, $S = -dE/dx$, but it is an experimental evaluation method rather than a separate formula from the manual. A polynomial fit to $E(x)$ was also used as a smooth visual guide, while the finite-difference values were used as the direct experimental stopping-power values.

3.8 SRIM Simulations

SRIM/TRIM simulations were used to compare the experimental results with simulated stopping powers and ranges. For the air-range comparison, helium ions with initial energy 5485.56 keV were simulated in dry air. For the Mylar comparison, helium ions with the same initial energy were simulated in a 40 μm Mylar target. The Mylar simulation used a target density of 1.397 g/cm^3 and 10000 simulated ions.

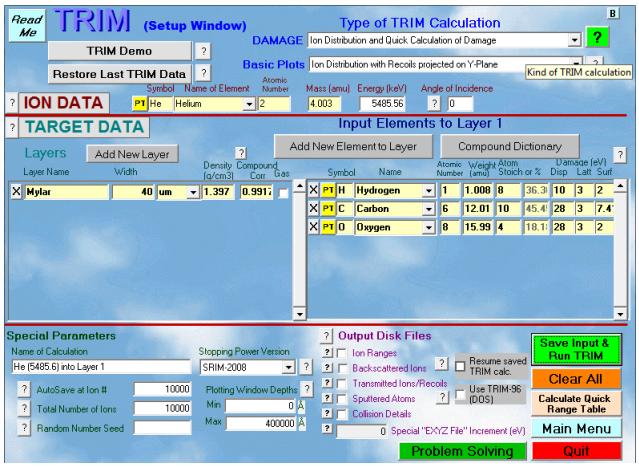


Figure 7: SRIM/TRIM setup window for the simulation of 5485.56 keV helium ions incident normally on a 40 μm Mylar target.

The extracted SRIM data gave the stopping power as a function of target depth, with separate ionization contributions from ions and recoils [1]. In the SRIM output, the depth is given in Ångström and the ionization energy units are given as:

$$\text{eV}/(\cdot\,\text{ion}). \tag{20}$$

Therefore, the SRIM data were converted using:

$$1\ \mu\text{m} = 10^4\ \text{Å}. \tag{21}$$

and

$$1\ \mathrm{eV/\AA} = 10\ \mathrm{keV}/\mu\mathrm{m}. \tag{22}$$

The simulated stopping-power curve was then plotted together with the experimentally determined finite-difference stopping-power values.

4. Results and Discussion

4.1 Finding Optimal Measurement Conditions

4.1.1 Measurement Procedure

The spectra of the ^{241}Am alpha source were recorded for different source–detector distances inside the vacuum chamber at a detector bias voltage of 50 V. The source position was varied using the available slot positions of the sample holder.

For each slot position, an alpha spectrum was recorded using the silicon surface barrier detector. The spectra were analyzed in the channel region containing the main alpha peak. To determine the peak width, a Gaussian fit was applied only to the right-hand side of the asymmetric peak profile. The fit region started slightly before the turning point of the peak maximum and extended over the complete high-energy tail region.

The full width at half maximum (FWHM) was extracted from the fitted Gaussian parameter using

$$\mathrm{FWHM} = 2\sqrt{2\ln 2}\,\sigma \tag{23}$$

where σ is the standard deviation of the Gaussian fit. The peak intensity was estimated from the maximum count value of the fitted peak. The normalized FWHM and normalized peak intensity were plotted as a function of slot number in order to determine the optimal measurement geometry.

4.1.2 Results

Figure 8 shows the measured alpha spectra for all slot positions together with the Gaussian fits applied to the right-hand side of each peak. The spectra show a decreasing count rate with increasing source–detector distance. At the same time, the peaks become narrower, indicating an improvement in energy resolution.

The extracted FWHM values and peak intensities are summarized in Table 2. Figure 9 shows the normalized FWHM and normalized count rate as a function of slot number.

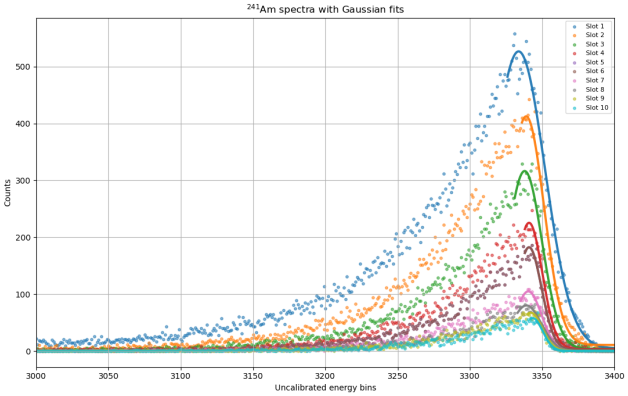


Figure 8: Representative alpha spectra measured at different source-detector distances.

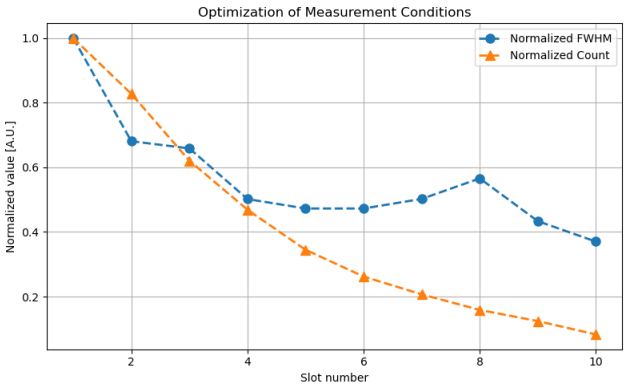


Figure 9: FWHM and Count vs Slot number plot

Table 2: Summary of distance-dependent peak properties.

Slot / Distance	$E_{\max}$	FWHM	Peak intensity
1	3338	42.84	550
2	3339	29.17	410
3	3340	28.22	315
4	3341	21.50	225
5	3341	20.25	190
6	3341	20.25	185
7	3342	21.52	105
8	3342	24.24	90
9	3343	18.57	70
10	3343	15.86	60

4.1.3 Discussion

The measured spectra show that the peak width decreases with increasing source–detector distance. At small distances, alpha particles emitted under large angles can still reach the detector. These particles traverse a larger effective thickness inside the source material and detector entrance region, resulting in increased energy loss and stronger energy straggling. Consequently, the observed peak becomes broader.

For larger distances, mainly particles emitted close to normal incidence reach the detector. The variation in path length is therefore reduced, which improves the energy resolution and decreases the FWHM.

However, increasing the distance also reduces the solid angle subtended by the detector, leading to a lower count rate. The normalized intensity decreases continuously with slot number, while the FWHM begins to stabilize after approximately slot 4 to slot 8. Therefore, slot 6 represents a suitable compromise between good resolution and sufficiently high counting statistics.

4.2 Spectrometer Calibration and Energy Resolution

4.2.1 Measurement Procedure

After the optimal measurement geometry was selected, a long spectrum of the ^{241}Am source was recorded at slot 6. The measurement was performed under vacuum with a detector bias voltage of 50 V and a recording time of approximately 15 min. The measured live time was 898 s.

The recorded spectrum was used to calibrate the channel axis of the multi-channel analyzer. The most probable alpha decay of ^{241}Am was used as the reference line,

$$E_\alpha = 5485.56\,\text{keV}. \tag{24}$$

The main alpha peak was fitted with a Gaussian function on the right-hand side of the asymmetric peak,

$$f(x) = A\exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right] + C. \tag{25}$$

The fitted peak position μ was assigned to the known alpha energy. Since only one calibration point was used, the intercept was set to zero and the channel width was calculated as

$$a = \frac{E_\alpha}{\mu}. \tag{26}$$

The calibrated energy scale was therefore obtained from

$$E(x) = ax. \tag{27}$$

The energy resolution of the detector was determined from the fitted full width at half maximum,

$$\text{FWHM} = 2\sqrt{2\ln 2}\,\sigma, \tag{28}$$

and

$$R = \frac{\text{FWHM}}{E_{\max}}. \tag{29}$$

For the dead-layer contribution, a SRIM/TRIM simulation was performed for helium ions with an initial energy of 5485.56 keV passing through a 50 nm silicon layer. The transmitted-ion file was used to obtain the energy distribution after the dead layer. The energy loss distribution was fitted with a Gaussian function, and its width was compared with the experimentally measured FWHM.

4.2.2 Results

Figure 10 shows the uncalibrated calibration spectrum together with the Gaussian fit. The fitted peak position was

$$\mu = (3341.03 \pm 0.17)\ \text{channels}. \tag{30}$$

This gives the channel width

$$a = (1.64187 \pm 0.00008)\ \text{keV/channel}. \tag{31}$$

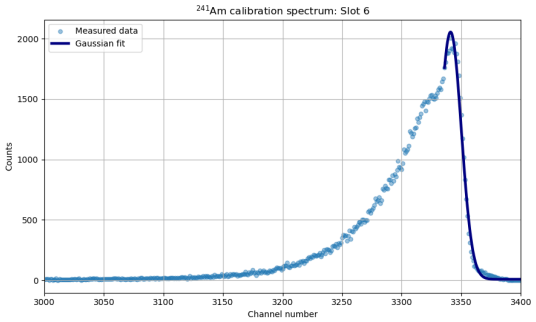


Figure 10: Uncalibrated ^{241}Am spectrum recorded at Slot 6 with a Gaussian fit to the high-energy side of the main alpha peak.

The corresponding calibrated spectrum is shown in Figure 11.

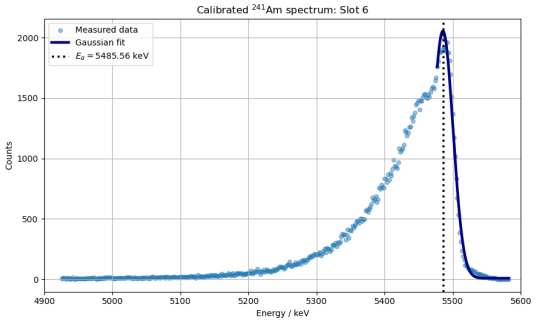


Figure 11: Calibrated ^{241}Am spectrum. The fitted peak position is assigned to the known alpha energy of 5485.56 keV.

Table 3: Summary of calibration and detector-resolution results.

Quantity	Value
Measurement slot	Slot 6
Live time	898 s
Reference energy E_α	5485.56 keV
Fitted peak channel μ	3341.03 ± 0.17
Channel width a	(1.64187 ± 0.00008) keV/channel
FWHM	(21.24 ± 0.45) channels
FWHM	(34.88 ± 0.74) keV
Energy resolution R	(0.0064 ± 0.0001)

The SRIM simulation result is shown in Figure 12. The transmitted alpha-particle energies were converted into energy losses inside the silicon dead layer.

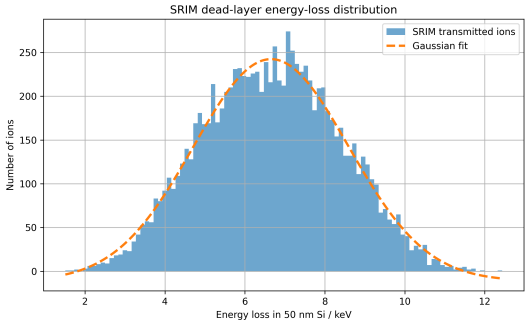


Figure 12: Histogram of the SRIM energy loss in the 50 nm silicon dead layer with Gaussian fit.

Table 4: Summary of SRIM dead-layer simulation.

Quantity	Value
Ion	Helium
Initial ion energy	5485.56 keV
Target material	Silicon
Dead-layer thickness	50 nm
Number of transmitted ions used	9999
Mean transmitted energy	5478.90 keV
Mean energy loss	6.65 keV
Standard deviation of energy loss	1.94 keV
$\mathrm{FWHM}_{\mathrm{straggling}}$	4.57 keV
$\mathrm{FWHM}_{\mathrm{straggling}}/\mathrm{FWHM}_{\mathrm{exp}}$	0.131

4.2.3 Discussion

The calibration spectrum recorded at Slot 6 shows the expected asymmetric alpha peak. This asymmetry is mainly caused by alpha particles losing different amounts of energy before reaching the active detector volume. Therefore, the Gaussian fit was applied to the high-energy side of the peak, which is less affected by the low-energy tail.

Using the most probable ^{241}Am alpha energy as a single reference point gives a channel width of approximately 1.642 keV/channel. The resulting experimental peak width is 34.88 keV, corresponding to an energy resolution of $R = 0.0064$. This value describes the relative width of the measured alpha peak compared with the alpha energy.

The SRIM simulation shows that the 50 nm silicon dead layer causes a mean energy loss of about 6.65 keV. More important for the resolution is the statistical fluctuation of this energy loss, commonly referred to as energy straggling. The simulated energy-loss distribution has a FWHM of only 4.57 keV, corresponding to approximately 13 % of the experimentally observed peak width. This indicates that the dead-layer straggling contributes only a small fraction of the total detector broadening. The dominant contribution to the measured FWHM must therefore originate from other effects such as electronic noise, charge-carrier statistics, incomplete charge collection, and additional energy losses in inactive detector regions. Consequently, the detector resolution is not limited by the dead layer itself but primarily by the intrinsic detector and readout system performance.

4.3 Range of Alpha Particles in Air

4.3.1 Measurement Procedure

To determine the range of alpha particles in air, spectra of the ^{241}Am source were recorded at different source-detector distances with air present in the chamber. The source position was varied from Slot 1 to Slot 10, corresponding to distances from 3 mm to 39 mm. In contrast to the calibration measurement, the energy loss in air was intentionally not suppressed, because the aim of this task was to measure how the alpha-particle energy decreases with increasing air path length.

For each distance, the main alpha peak was fitted using the Gaussian peak-fitting method introduced in Section 3.5.. The fitted peak channel μ was converted into an energy value E_{max} using the calibration relation in Eq. (12). The uncertainty of each converted energy value was calculated using the uncertainty propagation formula in Eq. (13).

The resulting values $E_{\mathrm{max}}(d)$ were plotted as a function of the source-detector distance d. The maximum range in air was then determined by fitting the measured data with a polynomial function and extrapolating the curve to the zero-energy condition defined in Eq. (5). The experimentally determined range was then compared with the empirical air-range relation from Eq. (6) and with the SRIM simulation.

4.3.2 Results

Fig:13 shows the measured spectra for the different source-detector distances in air. With increasing distance, the main alpha peak shifts steadily toward lower channels. This shift directly shows that the alpha particles lose kinetic energy while travelling through air before reaching the detector. In addition, the spectra recorded at larger distances are weaker and broader, especially close to the end of the alpha-particle range.

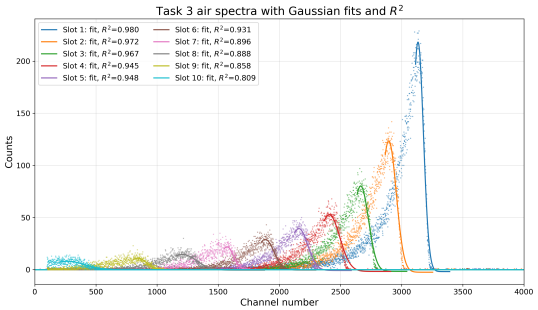


Figure 13: Alpha spectra recorded in air for different source-detector distances. The peak position shifts to lower channel numbers as the air path length increases.

The fitted peak positions and converted energies are listed in Table 5. The measured peak energy decreases from approximately 5145 keV at $d = 3$ mm to approximately 435 keV at $d = 39$ mm. This strong decrease confirms that

the alpha particles are already close to the end of their range at the largest measured distances.

Table 5: Distance-dependent alpha-particle energy in air. The peak positions were obtained from Gaussian fits to the measured spectra and converted to energy using the calibration factor $a = 1.64187\,\mathrm{keV/channel}$.

Slot	Distance d / mm	Peak channel μ	$E_{\max}$ / keV
1	3	3133.86 ± 0.62	5145.38 ± 1.05
2	7	2894.73 ± 0.94	4752.77 ± 1.56
3	11	2665.18 ± 1.30	4375.88 ± 2.15
4	15	2408.19 ± 1.37	3953.93 ± 2.26
5	19	2159.74 ± 2.34	3546.02 ± 3.85
6	23	1884.37 ± 2.51	3093.89 ± 4.12
7	27	1572.19 ± 2.40	2581.34 ± 3.95
8	31	1208.65 ± 3.36	1984.45 ± 5.53
9	35	842.25 ± 4.92	1382.86 ± 8.07
10	39	264.69 ± 14.91	434.58 ± 24.48

Fig:14 shows the extrapolation of $E_{\max}$ to zero energy. A cubic polynomial was used as an empirical interpolation function for the measured data. The zero-energy crossing, corresponding to the range condition in Eq. (5), gives the experimentally determined maximum range

$$R_{\mathrm{exp}} = (41.02 \pm 0.22)\ \mathrm{mm}. \tag{32}$$

The quoted uncertainty mainly describes the uncertainty of the polynomial extrapolation. It does not fully include all possible systematic effects, such as pressure and temperature deviations, finite slot-position uncertainty, or possible geometry-related uncertainties.

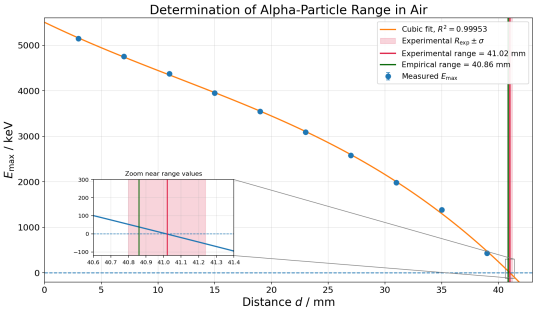


Figure 14: Determination of the maximum range of alpha particles in air by extrapolating $E_{\max}$ to zero. The experimentally determined range is compared with the empirical air-range prediction and the SRIM average range.

The SRIM ionization curve for alpha particles in air is shown in Fig:15. The curve shows the expected Bragg-curve behavior discussed in Section 2.3: the ionization energy loss increases as the particles slow down, reaches a maximum shortly before the end of the range, and then drops rapidly once the particles stop. This behavior is consistent with the stopping-power definition in Eq. (3) and with the qualitative Bragg-curve discussion in the theoretical background.

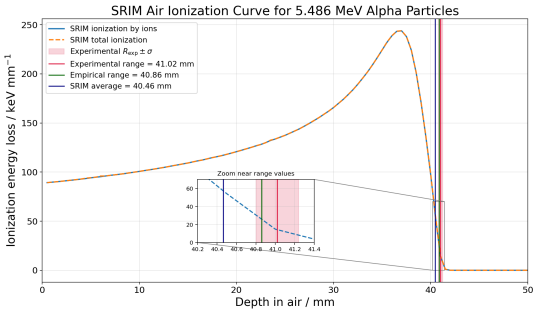


Figure 15: SRIM ionization curve for 5.486 MeV alpha particles in air. The ionization energy loss increases toward the end of the particle range and drops rapidly after the particles stop.

4.3.3 Comparison with Theory and SRIM

The empirical prediction was calculated using the air-range relation given in Eq. (6). For the dominant ^{241}Am alpha-particle energy,

$$E_\alpha = 5.48556\ \mathrm{MeV}, \tag{33}$$

this gives

$$R_{\mathrm{th}} = 40.86\ \mathrm{mm}. \tag{34}$$

The SRIM simulation gives an average range of

$$R_{\mathrm{SRIM}} = 40.46\ \mathrm{mm}, \tag{35}$$

with a range straggling of approximately

$$\sigma_{\mathrm{SRIM}} = 0.59\ \mathrm{mm}. \tag{36}$$

The comparison is summarized in Table 6. The relative deviation between the experimental result and the empirical formula is only 0.40%. The empirical prediction also lies within the quoted extrapolation uncertainty of the experimental result, since the difference between 41.02 mm and 40.86 mm is smaller than 0.22 mm. The deviation between the experimental result and the SRIM average range is larger, about 1.38%, but still comparable to the SRIM range straggling and to the expected systematic uncertainties of the measurement.

Table 6: Comparison of the experimentally determined alpha-particle range in air with the empirical prediction and the SRIM simulation.

Quantity	**Value**
Experimental range R_{exp}	(41.02 ± 0.22) mm
Empirical range R_{th}	40.86 mm
SRIM average range R_{SRIM}	40.46 mm
SRIM range straggling	0.59 mm
Deviation: experiment vs. empirical	0.40 %
Deviation: experiment vs. SRIM	1.38 %
Deviation: SRIM vs. empirical	0.96 %

4.3.4 Discussion

The measured decrease of $E_{\max}$ with increasing source-detector distance is the expected result for alpha particles travelling through air. Since alpha particles are heavy and doubly charged, they lose energy efficiently through ionization and excitation of air molecules. This is the physical process described by the stopping-power definition in Eq. (3). As the travelled distance increases, more energy is deposited in air before the particles reach the detector, so the remaining kinetic energy measured by the detector decreases.

The uncertainty of the fitted peak energy becomes larger at higher slot numbers, especially for Slot 10. This is mainly because the alpha particles are close to the end of their range. The detected peak is then weak, broad, and located at low energy, where the spectrum is less sharply defined. The Gaussian fit from Eq. (8) is therefore less constrained than for the high-energy spectra at short distances. In addition, the count rate decreases with increasing distance because the detector subtends a smaller solid angle. The reduced statistics further increase the uncertainty of the fitted peak position and therefore also the converted energy uncertainty calculated through Eq. (13).

The extrapolated experimental range agrees very well with the empirical range formula. The empirical value of 40.86 mm lies within the quoted experimental extrapolation uncertainty of $R_{\text{exp}} = (41.02 \pm 0.22)$ mm. This indicates that the measurement reproduces the expected range of ^{241}Am alpha particles in air. The SRIM average range is slightly lower than the experimental extrapolation, but the difference is still small. Possible reasons for the remaining deviation include pressure and temperature differences from the reference conditions of Eq. (6), finite uncertainty in the source-detector distances, imperfect peak fitting near the end of the range, and the fact that the extrapolation uncertainty does not include all systematic effects.

Overall, the air-range measurement is consistent with both the empirical prediction and the SRIM simulation. The result confirms the finite penetration depth of alpha particles in air and shows the characteristic increase in energy loss near the end of the range.

4.4 Energy Loss of Alpha Particles in Mylar Film

4.4.1 Measurement Procedure

The energy loss of alpha particles in Mylar was investigated by inserting Mylar foils of different thicknesses between the ^{241}Am source and the detector. The measurements were performed under vacuum so that the observed energy loss could be attributed mainly to the Mylar absorber rather than to energy loss in air.

All Mylar spectra were recorded with the source placed at Slot 8. Therefore, a separate no-Mylar reference spectrum recorded at Slot 8 was used as the zero-thickness reference. This ensured that the reference measurement and the Mylar measurements had the same source-detector geometry. The fitted energy of this reference spectrum was

$$E_0 = (5482.32 \pm 0.76)\ \text{keV}. \tag{37}$$

For each Mylar thickness, the main alpha peak was fitted using the Gaussian peak-fitting method described in Section 3.5. The fitted peak channel was converted to energy using the calibration relation in Eq. (12). The energy loss was then calculated using Eq. (17). The experimental stopping power was determined from neighboring thickness points using the finite-difference method given in Eq. (18), and each stopping-power value was assigned to the midpoint defined in Eq. (19).

4.4.2 Energy Loss Measurements

Figure 16 shows the alpha spectra recorded after transmission through the different Mylar thicknesses. As the Mylar thickness increases, the main alpha peak shifts toward lower channel numbers. This shift corresponds to a decreasing transmitted alpha-particle energy. At larger thicknesses, the peaks also become broader and weaker, indicating stronger energy straggling and reduced counting statistics near the end of the particle range.

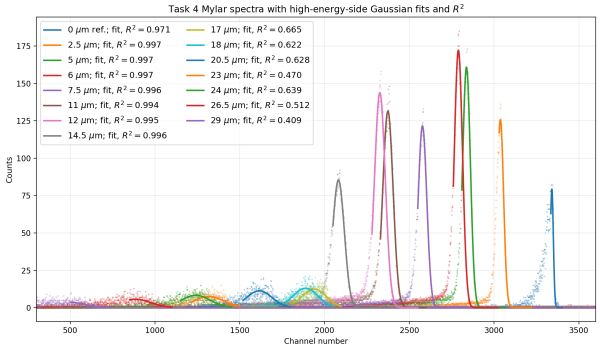


Figure 16: Alpha spectra measured after transmission through different Mylar thicknesses. The peak position shifts to lower channel numbers as the absorber thickness increases.

The fitted transmitted energies and calculated energy losses are summarized in Table 7. The transmitted energy decreases from 5482.32 keV without Mylar to 811.21 keV for a Mylar thickness of 29 μm. The corresponding energy loss increases to approximately 4671 keV.

Table 7: Measured transmitted alpha-particle energy and energy loss in Mylar. The zero-thickness value corresponds to the Slot 8 no-Mylar reference spectrum.

Thickness / μm	$E_{\max}$ / keV	ΔE / keV	Gaussian R^2
0	5482.32 ± 0.76	0.00	0.957
2.5	4982.61 ± 0.35	499.72	0.990
5	4657.11 ± 0.34	825.21	0.991
6	4578.42 ± 0.35	903.90	0.990
7.5	4230.06 ± 0.38	1252.26	0.988
11	3895.96 ± 0.43	1586.36	0.986
12	3817.56 ± 0.45	1664.76	0.986
14.5	3413.65 ± 0.49	2068.68	0.985
17	3179.78 ± 3.26	2302.54	0.718
18	3098.91 ± 3.50	2383.41	0.682
20.5	2637.93 ± 4.36	2844.39	0.588
23	2141.16 ± 5.95	3341.16	0.430
24	2003.55 ± 5.39	3478.78	0.541
26.5	1419.27 ± 8.82	4063.05	0.426
29	811.21 ± 11.65	4671.12	0.330

Figure 17 shows the transmitted energy as a function of Mylar thickness. The energy decreases nonlinearly with increasing absorber thickness. This nonlinearity is expected because the stopping power is not constant; according to the stopping-power definition in Eq. (3), the energy loss per unit path length depends on the particle energy.

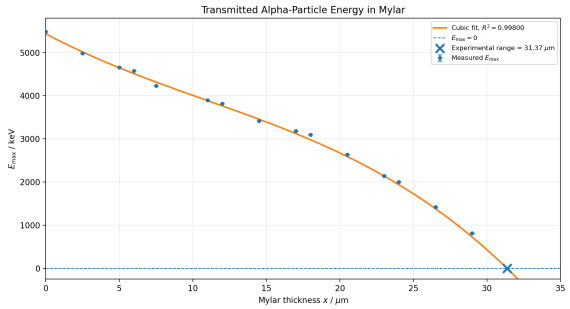


Figure 17: Transmitted alpha-particle energy as a function of Mylar thickness. The energy decreases nonlinearly as the alpha particles lose energy in the absorber.

The corresponding energy loss is shown in Figure 18. The energy loss increases continuously with Mylar thickness, reaching more than 4.6 MeV at the largest measured thickness.

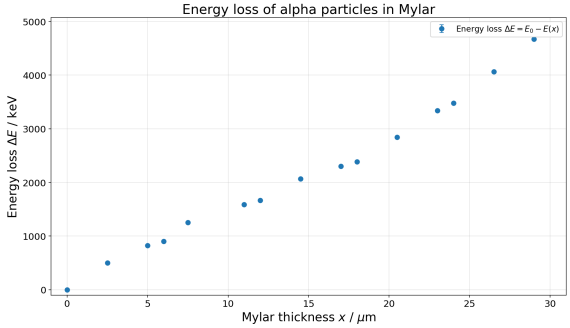


Figure 18: Energy loss of alpha particles as a function of Mylar thickness, calculated relative to the Slot 8 no-Mylar reference spectrum using Eq. (17).

4.4.3 Stopping Power

The experimental stopping power was calculated from neighboring energy values using the finite-difference approximation in Eq. (18). The resulting values are listed in Table 8 and plotted in Figure 19.

Table 8: Experimental stopping power of Mylar obtained from neighboring thickness intervals. Each value is assigned to the midpoint of the corresponding thickness interval.

Thickness interval / μm	Midpoint / μm	S / keV μm^{-1}
0–2.5	1.25	199.89 ± 0.34
2.5–5	3.75	130.20 ± 0.19
5–6	5.50	78.69 ± 0.49
6–7.5	6.75	232.24 ± 0.34
7.5–11	9.25	95.46 ± 0.16
11–12	11.50	78.40 ± 0.62
12–14.5	13.25	161.57 ± 0.27
14.5–17	15.75	93.55 ± 1.32
17–18	17.50	80.87 ± 4.78
18–20.5	19.25	184.39 ± 2.24
20.5–23	21.75	198.71 ± 2.95
23–24	23.50	137.61 ± 8.03
24–26.5	25.25	233.71 ± 4.14
26.5–29	27.75	243.23 ± 5.85

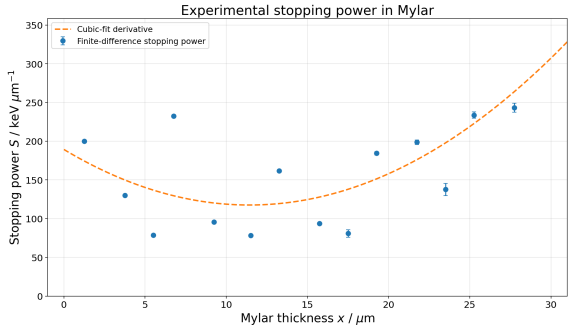


Figure 19: Experimentally determined stopping power of Mylar as a function of absorber thickness. The values were calculated using the finite-difference approximation in Eq. (18).

The stopping-power values show stronger scatter than the transmitted-energy values. This is expected because the finite-difference method uses differences between neighboring fitted energies. Small variations in peak position, foil thickness, or fit quality are therefore amplified. Nevertheless, the larger-thickness points show the expected tendency toward higher stopping power as the alpha particles slow down.

4.4.4 Comparison with SRIM

For comparison with simulation, a SRIM/TRIM calculation was performed for helium ions with initial energy 5485.56 keV incident normally on a 40 μm Mylar target. The simulation used a Mylar density of 1.397 g/cm^3 and 10000 ions. The extracted raw data gave the ionization energy loss as a function of depth in units of eV/(Å · ion) which were converted to keV/μm using the conversion factors in Eqs. (21) and (22).

The SRIM ionization curve is shown in Figure 20. The curve displays the characteristic Bragg behavior: the stopping power increases as the alpha particles lose energy, reaches a

maximum near the end of the range, and then drops rapidly once the particles stop.

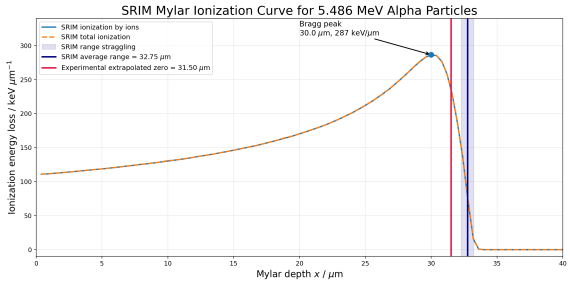


Figure 20: SRIM ionization curve for 5485.56 keV helium ions in Mylar. The stopping power reaches a maximum shortly before the alpha particles come to rest.

The experimental values are scattered, but they follow the same overall trend as the SRIM curve. In particular, the stopping power increases toward larger Mylar thicknesses, where the alpha particles have lower remaining kinetic energy.

The SRIM output gives an average alpha-particle range in Mylar of

$$R_{\mathrm{SRIM}} = 32.751\ \mu\mathrm{m}, \tag{38}$$

with range straggling

$$\sigma_{\mathrm{SRIM}} = 0.483\ \mu\mathrm{m}. \tag{39}$$

The cubic fit to the measured transmitted-energy curve gives an approximate zero-energy thickness of

$$R_{\mathrm{exp,Mylar}} \approx 31.50\ \mu\mathrm{m}. \tag{40}$$

The relative deviation between this extrapolated experimental Mylar range and the SRIM average range is approximately

$$\frac{|31.50 - 32.751|}{32.751} \cdot 100\% \approx 3.81\%. \tag{41}$$

The main SRIM comparison values are summarized in Table 9.

Table 9: Summary of the Mylar range and stopping-power comparison with SRIM.

Quantity	Value
Experimental extrapolated range in Mylar	31.50 μm
SRIM average range in Mylar	32.751 μm
SRIM range straggling	0.483 μm
Relative deviation: experiment vs. SRIM	3.81%
SRIM Bragg maximum position	$\approx$ 30.0 μm
SRIM maximum stopping power	$\approx$ 286.5 keV/μm

4.4.5 Discussion

The Mylar measurements show the expected decrease of transmitted alpha-particle energy with increasing absorber thickness. Since Mylar is much denser than air, the same alpha particles lose several MeV of energy over only a few tens of micrometers. This is consistent with the short range of alpha particles in solid materials.

The peak quality becomes worse for large Mylar thicknesses. This is visible in the decreasing Gaussian R^2 values in Table 7. For small thicknesses, the peaks are relatively sharp and the fits are reliable. For large thicknesses, especially above about 20 μm, the peaks become broader, weaker, and more asymmetric. This happens because the alpha particles have already lost most of their initial energy and experience stronger energy straggling. The lower count rate at large thickness also makes the fitted peak position less certain.

The experimental stopping-power values scatter around the SRIM curve rather than lying smoothly on it. This scatter is mainly caused by the finite-difference method. Since each stopping-power value is calculated from the difference between two neighboring fitted energies, even small errors in peak position or foil thickness can lead to a relatively large change in the calculated stopping power. This effect is especially strong for small thickness intervals, such as 5–6 μm or 11–12 μm.

Despite this scatter, the overall physical trend agrees with SRIM. The stopping power is not constant, but generally increases as the particles penetrate deeper into the Mylar and slow down. This agrees with the Bragg-curve behavior discussed in Section 2.3 and with the stopping-power definition in Eq. (3). The SRIM curve reaches its maximum near 30 μm, shortly before the simulated average range of 32.751 μm. The experimental extrapolated range of about 31.50 μm is close to this value, with a relative deviation of only 3.81%.

Possible sources of deviation between experiment and SRIM include uncertainty in the actual Mylar thickness, non-uniform foil stacking, small changes in source-detector alignment, limited counting statistics at large thickness, and the difficulty of fitting weak broadened peaks near the end of the alpha-particle range. In addition, the experimental stopping power is obtained from discrete thickness intervals, while SRIM provides a continuous depth-dependent stopping-power curve.

Overall, the Mylar measurement confirms the strong energy loss of alpha particles in solid absorber materials. The transmitted-energy curve, the increasing energy loss, and the comparison with SRIM all show behavior consistent with the expected slowing down and finite range of alpha particles in Mylar.

4.5 Summary of Uncertainties and Error Sources

Table 10: Main sources of uncertainty in the experiment.

Source of uncertainty	Expected influence
Peak fitting	Affects $E_{\max}$ and FWHM
Counting statistics	Affects peak intensity and fit uncertainty
Calibration uncertainty	Affects all converted energy values
Source-detector alignment	Affects intensity and resolution
Air pressure / temperature	Affects range in air
Mylar thickness uncertainty	Affects stopping power
Detector dead layer	Contributes to energy loss and broadening
SRIM assumptions	Affects simulation comparison

5. Conclusion

The measured dependence of the FWHM and count rate on the source–detector distance showed the expected trade-off between energy resolution and counting statistics. As discussed previously, the peak width decreased significantly for the first slot positions before beginning to stabilize, while the intensity continuously decreased with increasing distance. Slot 6 therefore provided a suitable compromise for the subsequent measurements by maintaining sufficiently good energy resolution together with acceptable counting statistics.

The detector calibration using the dominant ^{241}Am alpha line yielded an experimental energy resolution of approximately 0.64%, which is consistent with the expected performance of a silicon surface-barrier detector. The SRIM simulations additionally showed that a measurable fraction of the observed peak broadening originates from energy straggling inside the detector dead layer.

The experimentally determined alpha-particle range in air was found to be in very good agreement with both the theoretical estimate and the SRIM simulation within the expected experimental uncertainties. Similarly, the Mylar absorber measurements reproduced the expected increase of stopping power toward lower alpha-particle energies, consistent with the characteristic Bragg-like behaviour of charged particles in matter. Although the extracted stopping-power values exhibited noticeable fluctuations, particularly for larger absorber thicknesses, the overall trend remained physically reasonable and consistent with theoretical expectations.

Overall, the SRIM simulations showed good agreement with the experimental measurements for both the alpha-particle range and the energy straggling effects. The dominant limitations of the experiment originated from limited counting statistics at large absorber thicknesses, uncertainties in the asymmetric Gaussian peak fitting procedure, possible variations in the effective Mylar thickness, and systematic uncertainties associated with the single-point energy calibration. Nevertheless, the overall consistency between experiment, theoretical expectations, and SRIM simulations indicates that the obtained results are physically reasonable within the expected experimental uncertainties.

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Appendix

A.1 Reference Alpha Energy

The dominant alpha line of ^{241}Am was used for the one-point energy calibration. The value was taken from the reference data in the lab manual.

Table A1: Reference alpha-particle energy used for calibration.

Quantity	Value
Source isotope	^{241}Am
Dominant alpha energy E_α	5485.56 keV
Emission probability	84.8%
Daughter nucleus	^{237}Np
Use in report	Energy calibration reference

A.2 Raw Distance-Scan Data

Table A2 summarizes the spectra recorded under vacuum at different slot positions. These data were used to choose the optimal source-detector distance by comparing peak width and count rate.

Table A2: Raw peak properties from the vacuum distance scan used for optimizing the measurement geometry.

Slot	Peak channel	FWHM / channels	Peak intensity / counts
1	3338	42.84	550
2	3339	29.17	410
3	3340	28.22	315
4	3341	21.50	225
5	3341	20.25	190
6	3341	20.25	185
7	3342	21.52	105
8	3342	24.24	90
9	3343	18.57	70
10	3343	15.86	60

A.3 Calibration and Detector-Resolution Data

The calibration spectrum was recorded at Slot 6 under vacuum. The dominant ^{241}Am alpha peak was fitted on its high-energy side and assigned to the reference energy $E_\alpha = 5485.56$ keV.

Table A3: Calibration and detector-resolution parameters.

Quantity	Value
Measurement slot	Slot 6
Live time	898 s
Reference energy E_α	5485.56 keV
Fitted peak channel μ	3341.03 ± 0.17
Channel width a	(1.64187 ± 0.00008) keV/channel
FWHM	(21.24 ± 0.45) channels
FWHM	(34.88 ± 0.74) keV
Energy resolution R	0.0064 ± 0.0001

The Gaussian width corresponding to the fitted calibration peak can be obtained from

$$\sigma = \frac{\mathrm{FWHM}}{2\sqrt{2\ln 2}}. \tag{42}$$

Using the measured FWHM of (21.24 ± 0.45) channels gives

$$\sigma = (9.02 \pm 0.19) \text{ channels.} \tag{43}$$

A.4 SRIM Dead-Layer Simulation

A SRIM/TRIM simulation was used to estimate the contribution of the silicon dead layer to the observed peak broadening.

Table A4: Summary of the SRIM simulation for alpha-particle energy loss in the silicon dead layer.

Quantity	Value
Ion species	Helium / alpha particle
Initial ion energy	5485.56 keV
Target material	Silicon
Dead-layer thickness	50 nm
Number of transmitted ions used	9999
Mean transmitted energy	5478.90 keV
Mean energy loss	6.65 keV
Standard deviation of energy loss	1.94 keV
$\mathrm{FWHM}_{\mathrm{straggling}}$	4.57 keV
$\mathrm{FWHM}_{\mathrm{straggling}}/\mathrm{FWHM}_{\mathrm{exp}}$	0.131

A.5 Air-Range Data

Table A5 lists the fitted peak positions and converted energies used to determine the range of alpha particles in air.

Table A5: Distance-dependent alpha-particle energy in air.

Slot	Distance d / mm	Peak channel μ	E_{max} / keV
1	3	3133.86 ± 0.62	5145.38 ± 1.05
2	7	2894.73 ± 0.94	4752.77 ± 1.56
3	11	2665.18 ± 1.30	4375.88 ± 2.15
4	15	2408.19 ± 1.37	3953.93 ± 2.26
5	19	2159.74 ± 2.34	3546.02 ± 3.85
6	23	1884.37 ± 2.51	3093.89 ± 4.12
7	27	1572.19 ± 2.40	2581.34 ± 3.95
8	31	1208.65 ± 3.36	1984.45 ± 5.53
9	35	842.25 ± 4.92	1382.86 ± 8.07
10	39	264.69 ± 14.91	434.58 ± 24.48

Table A6: Comparison of the experimentally determined air range with the empirical prediction and SRIM.

Quantity	Value
Experimental range R_{exp}	(41.02 ± 0.22) mm
Empirical range R_{th}	40.86 mm
SRIM average range R_{SRIM}	40.46 mm
SRIM range straggling	0.59 mm
Deviation: experiment vs. empirical	0.40%
Deviation: experiment vs. SRIM	1.38%
Deviation: SRIM vs. empirical	0.96%

A.6 Mylar Energy-Loss Data

Table A7 lists the fitted transmitted energies and the corresponding energy losses for the Mylar absorber measurements. The zero-thickness value corresponds to the Slot 8 no-Mylar reference spectrum.

Table A7: Measured transmitted alpha-particle energy and energy loss in Mylar.

Thickness / μm	$E_{\max}$ / keV	ΔE / keV	Gaussian R^2
0	5482.32 ± 0.76	0.00	0.957
2.5	4982.61 ± 0.35	499.72	0.990
5	4657.11 ± 0.34	825.21	0.991
6	4578.42 ± 0.35	903.90	0.990
7.5	4230.06 ± 0.38	1252.26	0.988
11	3895.96 ± 0.43	1586.36	0.986
12	3817.56 ± 0.45	1664.76	0.986
14.5	3413.65 ± 0.49	2068.68	0.985
17	3179.78 ± 3.26	2302.54	0.718
18	3098.91 ± 3.50	2383.41	0.682
20.5	2637.93 ± 4.36	2844.39	0.588
23	2141.16 ± 5.95	3341.16	0.430
24	2003.55 ± 5.39	3478.78	0.541
26.5	1419.27 ± 8.82	4063.05	0.426
29	811.21 ± 11.65	4671.12	0.330

A.7 Mylar Stopping-Power Data

The stopping power was calculated by applying a finite-difference approximation to neighboring transmitted-energy values.

Table A8: Experimental stopping power of Mylar obtained from neighboring thickness intervals.

Thickness interval / μm	Midpoint / μm	S / keV μm^{-1}
0–2.5	1.25	199.89 ± 0.34
2.5–5	3.75	130.20 ± 0.19
5–6	5.50	78.69 ± 0.49
6–7.5	6.75	232.24 ± 0.34
7.5–11	9.25	95.46 ± 0.16
11–12	11.50	78.40 ± 0.62
12–14.5	13.25	161.57 ± 0.27
14.5–17	15.75	93.55 ± 1.32
17–18	17.50	80.87 ± 4.78
18–20.5	19.25	184.39 ± 2.24
20.5–23	21.75	198.71 ± 2.95
23–24	23.50	137.61 ± 8.03
24–26.5	25.25	233.71 ± 4.14
26.5–29	27.75	243.23 ± 5.85

A.8 Mylar SRIM Comparison

Table A9: Summary of the Mylar range and stopping-power comparison with SRIM.

Quantity	Value
Experimental extrapolated range in Mylar	31.50 μm
SRIM average range in Mylar	32.751 μm
SRIM range straggling	0.483 μm
Relative deviation: experiment vs. SRIM	3.81%
SRIM Bragg maximum position	$\approx$ 30.0 μm
SRIM maximum stopping power	$\approx$ 286.5 keV/μm

A.9 SRIM Simulation Parameters

Table A10 summarizes the main SRIM/TRIM simulations used in the analysis.

Table A10: SRIM simulation parameters used in the report.

Simulation	Ion energy	Target / medium
Detector dead layer	5485.56 keV	Silicon, 50 nm
Air range	5485.56 keV	Dry air, 5 cm
Mylar stopping power	5485.56 keV	Mylar, 40 μm, ρ = 1.397 g/cm^3

Table A11: Important SRIM output values used for comparison.

Quantity	Value
Number of ions in air simulation	10000
SRIM average range in air	40.463 mm
SRIM range straggling in air	0.588 mm
Number of ions in Mylar simulation	10000
Mylar target thickness	40.00 μm
Mylar density	1.397 g/cm^3
SRIM average range in Mylar	32.751 μm
SRIM range straggling in Mylar	0.483 μm
SRIM Bragg maximum in Mylar	$\approx$ 286.5 keV/μm at $\approx$ 30.0 μm

Use of AI Assistance

ChatGPT was used as an auxiliary tool during the preparation of this report. It was used to support the overall structure of the report, assist with wording and clarity, help identify inconsistencies in the draft, and provide guidance for data analysis and graph plotting. All experimental data, numerical calculations, plots, final interpretations, and conclusions were checked and verified by the authors. ChatGPT was not used as a primary scientific source for the physical theory or reference values.